

WILLINGNESS TO PAY FOR IMPROVED RESPIRATORY AND CARDIOVASCULAR HEALTH: A MULTIPLE-FORMAT, STATED-PREFERENCE APPROACH

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SUMMARY

This study uses stated-preference (SP) analysis to measure willingness to pay (WTP) to reduce acute episodes of respiratory and cardiovascular ill health. The SP survey employs a modified version of the health state descriptions used in the Quality of Well Being (QWB) Index. The four health state attributes are symptom, episode duration, activity restrictions and cost. Preferences are elicited using two different SP formats: graded-pair and discrete-choice. The different formats cause subjects to focus on different evaluation strategies. Combining two elicitation formats yields more valid and robust estimates than using only one approach.

Estimates of indirect utility function parameters are obtained using advanced panel econometrics for each format separately and jointly. Socio-economic differences in health preferences are modelled by allowing the marginal utility of money relative to health attributes to vary across respondents. Because the joint model captures the combined preference information provided by both elicitation formats, these model estimates are used to calculate WTP.

The results demonstrate the feasibility of estimating meaningful WTP values for policy-relevant respiratory and cardiac symptoms, even from subjects who never have personally experienced these conditions. Furthermore, because WTP estimates are for individual components of health improvements, estimates can be aggregated in various ways depending upon policy needs. Thus, using generic health attributes facilitates transferring WTP estimates for benefit–cost analysis of a variety of potential health interventions. Copyright © 2000 John Wiley & Sons, Ltd.

KEY WORDS — conjoint analysis; cost–benefit analysis; health valuation; stated preference; willingness to pay

INTRODUCTION

The economic analysis of many health intervention and regulatory programmes often requires evaluating the benefits of improved health. Obtaining credible measures of the economic value of morbidity, however, is one of the more difficult problems facing health economists. The existence of insurance, universal health care systems and

market participants that are unrepresentative of the population of policy interest often obscure essential supply and demand relationships. Thus, revealed-preference information generally has proven to be an insufficient basis for obtaining policy-relevant values of human health. In addition, studies that use contingent valuation (CV) are not well suited to valuing multiple health state attributes. Health economists increasingly are

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turning to stated-preference (SP) approaches as an alternative.^a

This study demonstrates the feasibility of applying SP techniques to elicit values for health conditions described in terms of symptom, activity restriction and duration. Furthermore, by combining two SP elicitation methods, graded-pair and discrete-choice, the estimates presented in this study are more valid and robust measures of benefits than could be obtained from a single format. Our results demonstrate sensitivity to the symptoms being considered as well as to the activity restrictions associated with those symptoms. Subjects clearly indicate systematic preferences for milder morbidity effects over more severe ones. These preferences translate into willingness to pay (WTP) estimates that vary logically with severity.

SP methods evolved as market research tools for evaluating consumer behaviour and predicting sales of new products (Cattin and Wittink [1]; Wittink and Cattin [2]). SP recently has been applied in environmental and health economics as an alternative to CV methods.^b Viscusi *et al.* [3] and Krupnick and Cropper [4] (using the Viscusi data) use SP analysis to elicit a value from subjects for reducing chronic health risks. Other SP studies have elicited preferences for other health and health care attributes, including Hakim and Pathak [5], Vick and Scott [6], Ryan *et al.* [7]; Ryan and Hughes [8], Bryan *et al.* [9], Van der Pol and Cairns [10], Propper [11] and Farrar and Ryan [12].

In this study, specified health states consist of multiple attributes. We presume that people have preferences for different levels within these attributes and are willing to accept some trade-offs among them.^c The utility obtained under different health states is derived from the revealed trade-offs.^d By including a cost attribute, we can use the implicit marginal utility of money to scale changes in health state utility in monetary units.

GRADED-PAIR AND DISCRETE-CHOICE SP QUESTIONS

Two types of SP analysis lend themselves to valuing health effects.^e The first, *graded pairs*, measures subjects' valuations of variations in attributes by requiring them to evaluate trade-offs among various attributes. In a graded-pair for-

mat, subjects sequentially are presented with several different pairs of bundled commodities, represented as sets of attribute levels, and asked to compare each pair. They are asked to rate the intensity of their preference for one of the pairs on a numerical scale, say from 1 to 7, where 1 indicates a strong preference for the first bundle, 7 indicates a strong preference for the second bundle and 4 indicates indifference between the two bundles. The subject is asked to rate a series of these pairs, with each pair having different attributes or attribute levels. (See Viscusi *et al.* [3] for an example of this approach in valuing bronchitis risks.)

Discrete choice, in contrast, confronts subjects with several different products or programmes simultaneously and simply asks them to identify the most-preferred alternative in the choice set.^f As with the graded-pair approach, each commodity is described as a set of attributes. However, the subjects do not provide a rating of the intensity of their preferences. Typically, each subject is shown a series of the choice sets to evaluate.^g

Huber *et al.* [13] argue that each format has relative merits because each relies on a different cognitive process for eliciting preferences.^h The different formats cause subjects to focus on different evaluation strategies. Huber *et al.* [13] cite a well-established result in marketing that the choice format is a better predictor of short-term market behaviour in familiar, repeat-purchase situations, while graded-pairs is a better predictor of longer-term market adaptations to new products with relatively unfamiliar attributes. The health outcomes analysed in this study have attributes of both kinds of situations in different degrees for different subjects. Furthermore, real-world behaviour often involves a mixture of valuation strategies. Thus, we have no *a priori* basis for choosing between graded-pairs or choice-format as the better measure of health preferences in this context. Huber *et al.* [13] have found that employing more than one elicitation format may predict behaviour better than using one format alone, simply because each format requires different heuristics and may provide only a partial picture of preferences. In this study, therefore, we have adopted this approach and apply both graded-pair and discrete-choice formats to estimate joint WTP estimates. In other situations, one or the other approach might be more appropriate.

Table 1 shows the levels associated with each attribute in the two SP formats. We used a simplified Quality of Well Being (QWB) framework for defining health attributes for this study. The QWB health classification system defines health states in four dimensions: three function states (mobility, physical activity and social activity) and the most severe symptom/problem complex. Instead of using the three function states as separate attributes, we combined them into one attribute called 'daily activity'.ⁱ We then added the attributes of duration and annual subject costs to

characterize the desired health state. Pre-tests indicated that higher cost levels were required in the choice format to induce sufficient trade-offs.^j

The experimental design consists of main effects, nearly orthogonal arrays of 40 graded pairs and 40 choice sets using the attribute levels in Table 1. Zwerina *et al.* [14] describe a search procedure for identifying a nearly optimal choice-experiment design. We adapted their algorithm, allowing for the necessity of imposing constraints that eliminate implausible or uninformative combinations of levels and pairs (see Johnson *et al.*

Table 1. Attribute and attribute levels shown in graded-pair and discrete-choice comparisons

Attribute	Level	Description
Symptom	Nose	Stuffy/runny nose and sore throat
	Eye	Eye irritation
	Flutter	Fluttering in chest and feeling light-headed
	Breath	Coughing, wheezing, shortness of breath
	Ache	Coughing or wheezing with fever, chills or aching all over
	Swell	Shortness of breath, and swelling in ankles and feet
	Pain	Pain in chest or arm
Duration	1	5
	1-day episode	5-day episode
		10
Daily activity	NOLIM	You can go to work, go to school, do housework, participate in social or recreational activities and have no physical limitations
	SOMELIM	You can go to work, go to school, do housework and participate in social or recreational activities, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition
	NOSOC	You can go to work, go to school, do housework, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) and cannot participate in social or recreational activities because of this health condition
	ATHOME	You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition, but you can care for yourself
	NEEDHELP	You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you need help caring for yourself (feeding, bathing, dressing, toilet) because of this health condition
	INHOSP	You are in hospital and need help caring for yourself (feeding, bathing, dressing, toilet)
Annual costs	Graded-pair (CAN\$)	Discrete-choice (CAN\$)
	10	50
	25	100
	50	200
	100	300
	200	500
	500	750

[15]).^k For example, the relatively mild symptoms 'Stuffy/runny nose and sore throat' (NOSE) and 'Eye irritation' (EYE) are never seen with activity restrictions greater than 'Social and recreation limitations' (NOSOC), nor with costs greater than CAN\$100. Similarly, more severe conditions are never seen with the mildest activity restriction level, NOLIM.^l Given subjects' practical time and attention constraints, we administered eight graded pairs and eight choice sets to each subject by computer, each set of which was drawn randomly from a five-block design of graded pairs and a five-block design of choice sets.

Figure 1 illustrates an example of a graded pair question. In this example, the price is expressed as illness-related costs.^m The subjects indicate their preferences for Condition A versus Condition B. Because there is no unambiguous baseline reference point in the graded-pair questions and no health conditions with zero cost, the graded-pair questions are designed to obtain information on marginal trade-offs among health attributes and costs.

Marketing applications of choice formats generally allow subjects to choose 'current brand' or 'no purchase' (Olsen and Swait [16]; Ruby *et al.* [17]). Including such an opt-out alternative ensures realistic representation of consumer choice, provides an unambiguous reference point from

which to calculate WTP and allows calculation of market shares among competing alternatives. It is not clear what the corresponding opt-out alternative is in a health context. Patients generally do not choose to be sick. Rather, they are endowed with health states and have some control over treatment options. 'No purchase' presumably means remaining in a less-desirable health state for a longer period of time. Returning to normal or good health may require substantial health care expenditures. Thus, we have adapted the market goods choice format to realistically represent health care choices. Current health is the reference point for welfare calculations. Thus, WTP is defined as the monetary equivalent of the utility difference between current health and a specified state of ill health.

Figure 2 illustrates the discrete-choice format used in this study.ⁿ The discrete-choice format directly elicits total values for movements from a given diminished health state to the subject's current health state. The left-hand profile consists of a relatively severe hypothetical initial condition with zero cost. Alternatives A and B represent two courses a subject could choose if experiencing the initial condition. Alternative A portrays a condition of intermediate severity and intermediate cost. Alternative B is described as the subject's current health state on the day of the survey with

Category	Condition A	Condition B
Duration of Episode	5 days	1 day
Symptoms	Eye irritation (itching, burning, redness)	Pain in chest or arm
Daily Activities	• CAN go to work, go to school, do housework • CANNOT participate in social or recreational activities • Have SOME physical limitations • CAN care for yourself	• CANNOT leave your house, go to work, go to school, do housework, and participate in social or recreational activities • Have SOME physical limitations • CAN care for yourself
Total costs of this episode to your household	\$50	\$200

1 2 3 4 5 6 7
 A is A is A is A and B is B is B is
 much somewhat slightly A and B slightly somewhat much
 better better better are about better better better

Please press a number from 1 to 7 that best reflects your rating.

Figure 1. Example of graded-pair question

1	2	3
Prefer Initial Condition	Prefer Alternative A	Prefer Alternative B

Please press a number from 1 to 3 that best reflects your choice if faced with these options.

a relatively high cost. In essence, subjects can choose to remain in a relatively severe condition and not pay any additional costs for treatments outside the government health plan or insurance plan, or they can choose to pay for additional treatments to improve their health to their current health state.

both of the hypothetical alternatives. In such cases, at least one of the options would be preferred, or dominant, in all of the attributes, and thus choices would reveal less information about trade-off relations. This situation does not invalidate the design, but introduces some degree of statistical inefficiency.

The health history questions collect information on the subject's personal health history, the health history of family members and socio-demographic information about the subject and the subject's household. The information gathered is used in

the analysis to determine whether health history and other demographic variables influence WTP for health. This information also was used to characterize the subject's current health condition in Alternative B of the choice sets. The instrument was administered to 399 randomly recruited subjects in the Toronto area between March and July 1997.⁹

ANALYSIS OF GRADED-PAIR DATA

Graded-pair responses are ordinal ratings of utility differences between attribute level pairs. Estimation strategy should thus account for the discrete, ordinal nature of the response variable. In this section, we describe a general model for estimating subject utility functions from such data and a procedure for calculating marginal WTP using the estimated utility functions.^r We assume that individual indirect utility can be expressed as a function of commodity attributes and personal characteristics:

$$U_{st}^i = V^i(X_{st}, Z^i, P_{st}; \beta^i, \delta^i) + e_{st}^i \quad (1)$$

where U_{st}^i is individual i 's utility for commodity profile st , where $s = L$ or R , denoting the left side and right side profiles for pair t , $t = 1, \dots, 8$; $V^i(\cdot)$ is the non-stochastic part of the utility function; X_{st} is a vector of attribute levels in profile st ; Z^i is a vector of personal characteristics; P_{st} is the cost of the commodity profile; β^i is a vector of attribute parameters; δ^i is the marginal utility of money; and e_{st}^i is a disturbance term.

The utility difference for profile pair t , dU_t^i , is simply

$$dU_t^i = V_{Rt}^i - V_{Lt}^i + \varepsilon_t^i \quad (2)$$

where V_{Rt}^i and V_{Lt}^i are the indirect utilities associated with the right side and left side profiles, respectively, and $\varepsilon_t^i = e_{Rt}^i - e_{Lt}^i$ is the associated disturbance term. The disturbance term captures the effects of unobserved factors, including possible inherent ambiguity of subject preferences and cognitive errors.

The difference in indirect utility for commodity pair t , dU_t^i , often is specified as a simple linear function of attributes

$$\begin{aligned} dU_t^i &= V_{Rt}^i - V_{Lt}^i + \varepsilon_t^i \\ &= \left[\sum_h \beta_h \cdot X_{hRt} + \delta \cdot P_{Rt} \right] \\ &\quad - \left[\sum_h \beta_h \cdot X_{hLt} + \delta \cdot P_{Lt} \right] + \varepsilon_t^i \end{aligned} \quad (3)$$

where h indexes attributes. This specification assumes that attributes are neither substitutes nor complements for each other, so a change in the level of one attribute does not affect the marginal utility of any other attribute.^s

The effects of personal characteristics on utility differences do not appear in Equation (3) because subject personal characteristics do not vary between the left and right sides of the screen. Controlling for such variables requires interacting them with commodity attributes or prices that do vary between profiles. For example, we can estimate the marginal utility of money as a function of individual characteristics Z^i and employ a functional form that allows for diminishing marginal utility of money.

$$\begin{aligned} dU_t^i &= V_{Rt}^i - V_{Lt}^i + \varepsilon_t^i \\ &= [X_{Rt} \cdot \beta + (Z^i \cdot \gamma) \cdot \sqrt{P_{Rt}}] \\ &\quad - [X_{Lt} \cdot \beta + (Z^i \cdot \gamma) \cdot \sqrt{P_{Lt}}] + \varepsilon_t^i \end{aligned} \quad (4)$$

where γ is a vector of marginal utility of money parameters.^t

Similarly, the β coefficients on attributes could also be allowed to vary across subjects by making them functions of individual characteristics. It is also possible to employ more general non-linear specifications of continuous attribute variables or interactions among attribute variables.

We do not observe dU_t^i directly. Instead, we observe C_t^i , which is a discrete rating category related to the unobserved dU_t^i of interest. The appropriate approach, therefore, is ordered logit or probit, which incorporates both the discreteness and the natural ordering of the data. This study uses ordered probit, which assumes the error term is normally distributed. To estimate ordered probit models, the data are sorted so that the preferred profile is on the right, making $dV_t^i = V_{Rt}^i - V_{Lt}^i \geq 0$.^u We construct the rating categories by recoding responses accordingly, so that 0 indicates indifference and 3 indicates maximum difference.^v Because probit assumes the Equation (3) error term ε_t^i is distributed $N(0, \sigma^2)$, the probability of observing response C_t^i is

$$\text{Prob}(C_t^i = k) = \Phi[\mu^i \cdot (\alpha_k - dV_t^i)] - \Phi[\mu^i \cdot (\alpha_{k-1} - dV_t^i)], \quad k = 0, 1, \dots, 3, \quad (5)$$

where Φ is the cumulative standard normal distribution function, the α_k are threshold constants, and scale parameter μ^i is the inverse of the standard deviation.^w

Because the health valuation survey collects eight responses from each subject, it is appropriate to estimate a panel model, which accounts for correlated errors in each subject's series of ratings. A random effects model incorporates an individual-specific error term, so that

$$\varepsilon_t^i = \eta_t^i + \lambda^i, \quad (6)$$

where η_t^i is a common error term distributed by $N(0, \sigma_\eta^2)$, λ^i is an individual-specific error term distributed by $N(0, \sigma_\lambda^2)$ and $\sigma_\varepsilon^2 = \sigma_\eta^2 + \sigma_\lambda^2$. The correlation between successive disturbances for the same subject is ρ .

Most ordered-category data contain no information on how variance or scale might vary across subjects, and thus it is usually normalized uniformly to one. However, graded-pair data include both cross-section and time-series information, thus it is possible to obtain scale estimates. We can account for non-uniform scale by letting ε_t^i vary across subjects so that $\varepsilon_t^i \sim N(0, 1/\mu_i^2)$.^x Alternatively, we can make μ_i a systematically varying parameter that is a function of personal characteristics Y^i , such as income and education, so that $\varepsilon_t^i \sim N[0, 1/\mu(Y^i)^2]$.

Integrating out the random effect, Equation (5) now becomes

$$\begin{aligned} \text{Prob}(C^i = k_1^i, k_2^i, \dots, k_8^i) \\ = \int_{-\infty}^{\infty} \phi(\lambda^i) \cdot \prod_{t=1}^8 \Phi[\mu^i \cdot (\alpha_{k_t} - dV_t^i)] \\ - \Phi[\mu^i \cdot (\alpha_{k_t-1} - dV_t^i)] d\lambda^i, \end{aligned} \quad (7)$$

where $\phi(\lambda^i)$ is the normal probability density function for λ^i .^y

Estimating individual scale parameters provides a means of quantifying the coherence of individual rating patterns and of controlling for different levels of variance in subjects' error terms. Subjects who refuse to solve or have difficulty solving the utility difference problem and who enter random or repetitive ratings will have unusually noisy ratings and thus larger estimated variance. Scale estimates thus provide a means of testing whether groups of subjects are experiencing problems with

the survey design. The consistency of a subject's rating pattern can vary by the degree of attentiveness, quantitative orientation, age, educational background, susceptibility to fatigue, and other factors. In addition, we can identify individual subjects who may have failed to perform the rating task properly. Finally, individual scale estimates help account for differences in how subjects interpret and use the 0-to-3 rating categories. Some subjects may be reluctant to use the entire range of ratings, which is a common problem in graded-pair elicitations [18,19].

ANALYSIS OF DISCRETE-CHOICE DATA

The SP survey also included a series of choice judgements with three alternatives in each choice set. The linear specification of utility analogous to Equation (4) for the three alternatives is

$$\begin{aligned} U_{jt}^i &= V_{jt}^i + \varepsilon_{jt}^i = X_{jt} \cdot \beta + (Z^i \cdot \gamma) \sqrt{P_{jt}} + \varepsilon_{jt}^i, \quad j = 1, 2; \\ U_{jt}^i &= V_0^i + \delta_t^i + \varepsilon_{0t}^i = H^i \cdot \omega + (Z^i \cdot \gamma) \sqrt{P_{jt}} + \varepsilon_{jt}^i, \\ &j = 3, \end{aligned} \quad (8)$$

where U_{jt}^i , $j = 1, 2$, is the utility of each of the two alternative health profiles, and U_{jt}^i , $j = 3$, is the utility of the current health alternative. V_0^i is the subject's status quo health state utility, H^i is a vector of health history variables and ω are associated parameters. Other variables were defined previously.

Assuming ε follows a type 1 extreme value error structure, the probability that alternative j will be selected from choice set t is the standard conditional-logit expression

$$\text{Prob}(C_t^i = j) = \frac{\exp(\mu^i \cdot V_{jt}^i)}{\sum_{k=1}^3 \exp(\mu^i \cdot V_{kt}^i)}, \quad (9)$$

where C_t^i is the selected alternative in each of eight choice sets, μ^i is the scale parameter and V_j^i is the determinate part of the utility of alternative j .^z Note that individual characteristics again fall out of this expression unless interacted with health attributes. The scale parameter μ^i also is generally not identifiable in such models and is normalized at 1. However, because we have

multiple observations for each subject, we can estimate $\mu^i = \mu(Z^i)$. Like the ordered-probit analysis described above, the conditional logit model specified by Equations (8) and (9) is estimated using maximum likelihood.^{aa}

Conditional logit models are known to be subject to violations of the irrelevance of independent alternatives assumption (IIA). This condition requires that the ratio of probabilities for any two alternatives be independent of the attribute levels in the third alternative. Tests of the choice data indicate that conditional logit estimates do not satisfy this requirement. Under these conditions, parameter estimates are biased. Second, the conditional logit models assume that differences in WTP across subjects arise only from differences in the marginal utility of money. Finally, conditional logit does not account for correlations within each subject's series of choices. Revelt and Train [20] recently have proposed using random parameter logit (RPL) for SP data similar to ours. RPL is not subject to the IIA assumption, accommodates correlations among panel observations and accounts for uncontrolled heterogeneity in tastes across subjects.

Modifying Equation (8) to introduce subject-specific stochastic components for each β

$$U_{jt}^i = V_{jt}^i + \varepsilon_{jt}^i \equiv X_{jt} \cdot (\beta + \eta^i) + (Z^i \cdot \gamma) \sqrt{P_{jt}} + \varepsilon_{jt}^i. \quad (10)$$

Equation (9) now becomes

$$\text{Prob}[C^i = (C_{j1}^i, C_{j2}^i, \dots, C_{j8}^i)]$$

$$= \prod_{t=1}^8 \left[\frac{\exp[\mu^i \cdot V_{jt}^i(\beta^*)]}{\sum_{k=1}^3 \exp[\mu^i \cdot V_{kt}^i(\beta^*)]} \right] \quad (11)$$

where now $\beta^* = (\beta + \eta^i)$. In contrast to conditional logit, the stochastic part of utility now may be correlated among alternatives and across the sequence of choices via the common influence of η^i . McFadden and Train [21] show that any random utility model can be approximated by some RPL specification.

CALCULATING WILLINGNESS TO PAY

Estimating the parameters of the utility function enables us to quantify the value of changes in commodity attributes. The units of the estimated

utility index essentially are arbitrary. However, this arbitrary measure can be converted into a dollar metric using the estimated marginal utility of money. Let X_j^0 indicate the status quo vector of attribute levels. In our case, this corresponds to the subject's current health state. The cost of X_j^0 is P_j^0 , which we take to be zero. X_j^* indicates a changed vector of attribute levels corresponding to a given combination of symptom and activity level. The WTP for a given change in commodity attributes ($X_j^* - X_j^0$) is the amount of money ($P_j^* - P_j^0$) that would leave subject i indifferent between the payment and the change in attribute levels, so that P_j^* satisfies

$$\begin{aligned} V^i[X_j^*, Z^i, P_j^*; \beta^i, \delta^i(P, Z^i)] \\ = V^i[X_j^0, Z^i, P_j^0; \beta^i, \delta^i(P, Z^i)]. \end{aligned} \quad (12)$$

In the linear specification of Equations (4) and (10), the β coefficient for each attribute represents its constant marginal utility. The negative of the coefficient of the price attribute is interpreted as the marginal utility of money, or the utility derived from more dollars. Thus,

$$\begin{aligned} \text{WTP}^i(X_j^* - X_j^0) &= \frac{\sum_h (X_{hj}^* - X_{hj}^0) \frac{\partial V^i}{\partial X_{hj}}}{-\frac{\partial V^i}{\partial P}} \\ &= \frac{\sum_h (X_{hj}^* - X_{hj}^0) \beta_h^i}{-\frac{(Z^i \cdot \gamma)}{(2 \cdot \sqrt{P})}}, \end{aligned} \quad (13)$$

where h indexes attributes (McFadden [22]).

Allowing the marginal utility of money to vary with personal characteristics allows for differences in tastes to affect the relative utilities of health and money, even though the β_j parameters are constant across subjects. This specification facilitates transferring WTP estimates to populations with different demographic characteristics than our sample. Thus, because WTP estimates are obtained by dividing the health coefficients by the marginal utility of money, changing values of the personal characteristics included in the utility of money function affects the WTP estimates.

Any payment less than or equal to $\text{WTP}^i(X_j^* - X_j^0)$ leaves individuals at least as well-off as they would be if the change ($X_j^* - X_j^0$) had not occurred. Because we recover the parameters for a complete utility index, WTP can be constructed for any utility difference, assuming that the

marginal utility of money is constant for all utility differences of interest.^{bb}

GRADED-PAIRS ESTIMATES

Table 2 contains the definitions of the variables used in all analyses in this study. Attributes are effects-coded rather than dummy variable coded.^{cc} Consequently, the omitted categories NOSE and NOLIM are the negative sum of the included symptoms, interacted with the log of duration, and activity categories.^{dd} The symptom coefficients represent the disutility of the interacted symptom relative to the mean effect. The daily activity restriction levels represent the additional disutility of experiencing these restrictions compared with experiencing no limitations (NOLIM). We report both nominal and normalized coefficients. The normalized coefficients rescale the health index from 0 to 1, with the INHOSP coefficient equal to zero (the least utility) and the NOLIM coefficient equal to one (the most utility). The normalized coefficients facilitate comparisons of marginal utility across the symptom and activity levels within this model.

Table 3 reports estimates for the graded-pair, ordered-probit panel model.^{ee} This model estimates and controls for the correlation among the subject's eight SP answers due to unobserved heterogeneity across subjects. The correlation coefficient, RHO, shows that there is significant correlation among the eight SP questions. Controlling for this correlation affects the significance of several variables.^{ff}

Symptom attribute levels are interacted with the log of duration to facilitate comparisons with the discrete-choice estimates.^{gg} There is no natural ordering of symptom disutility, so we have no specific expectations about relative magnitudes. Nevertheless, it is not surprising that the coefficients on NOSE and EYE indicate higher utilities than the other symptoms, and these differences are significant at the 1% level.^{hh} Overall, there is relatively little variation in the normalized symptom coefficients compared with the activity level coefficients, as the normalized symptom coefficients range from 0.3535 to 0.6096. However, five of the seven symptoms are significantly different than the overall mean symptom effect. In addition, PAIN and BREATH have significantly less disutility than FLUTTER, ACHE and SWELL.

Unlike the symptom variables, the activity restriction levels have a natural ordering with 'no limitations' (NOLIM) representing the best activity restriction level and 'in hospital' (INHOSP) representing the worst activity restriction level. All of the coefficients on the activity restriction levels are significantly different than the mean activity effect. The coefficients of SOMELIM and NOSOC are reversed in order but are statistically equivalent, indicating that subjects did not have strong preferences for not being able to participate in social and recreational activities and having physical limitations versus simply having physical limitations. The remaining coefficients decrease monotonically, as expected, indicating that greater activity restrictions result in significantly greater utility losses.

Allowing the marginal utility of money δ^i to vary with personal characteristics allows for differences in tastes to affect the relative utilities of health and money, even though the β_j parameters are constant across subjects. The utility of money is the inverse of the disutility of cost. Thus, a positive coefficient on a covariate means that an increase in the variable moves the disutility of costs closer to zero, resulting in higher WTP. Alternatively, a negative coefficient results in lower WTP estimates. Except for VIMPORTANT and EDUCATION, the utility of money covariates are significant, suggesting that different groups of subjects have different preferences for health state/money trade-offs. We did not have specific expectations of the sign of SCORE. However, the results show that subjects with higher quiz scores have lower WTP for improved health, suggesting that subjects who accurately processed the information in the article are willing to pay less to avoid the specific health effects discussed. On the other hand, older and symptomatic subjects have higher WTP, possibly because their current health state is poor and thus marginal improvements are more valuable or because they want to avoid the recurrence of an illness.

The scale variables measure the amount of variance in the error term, indicating and controlling for the amount of 'noise' in the subject's ratings. Because the scale factor is inversely related to the variance of the error term, as the scale factor increases, the variance decreases. The significance of the scale parameters indicates that subjects used the rating scale differently. Of the significant coefficients in the scale function, subjects with

Table 2. Variables used in analysis

Variables	Description
NOSE	Stuffy or runny nose and sore throat
EYE	Eye irritation
FLUTTER	Fluttering in chest and feeling light-headed
BREATH	Coughing, wheezing, shortness of breath
ACHE	Coughing or wheezing with fever, chills or aching all over
SWELL	Shortness of breath and swelling in ankles and feet
PAIN	Pain in chest or arm
LNDAYS	Log of the number of days of episode (1, 5, 10) plus one
NOLIM	Activity level 1: You can go to work, go to school, do housework, participate in social or recreational activities and have no physical limitations
SOMELIM	Activity level 2: You can go to work, go to school, do housework and participate in social or recreational activities, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition
NOSOC	Activity level 3: You can go to work, go to school, do housework, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) and cannot participate in social or recreational activities because of this health condition
ATHOME	Activity level 4: You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition, but you can care for yourself
NEEDHELP	Activity level 5: You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities and you need help caring for yourself (feeding, bathing, dressing, toilet) because of this health condition
INHOSP	Activity level 6: You are in hospital and need help caring for yourself (feeding, bathing, dressing, toilet)
SQTCOST	Square root of the cost levels per year (CAN\$10–CAN\$750)
SCORE	Quiz score (percent correct)
AGE	The midpoint of the age category
AGESQUARED	The square of AGE
EDUCATION	Number of years of education
SYMPTOMATIC	Dummy variable = 1 if subject has ever been diagnosed with any cardiovascular or respiratory conditions or other serious illnesses
VIMPORTANT	Number of health factors subjects indicated as being very important to a person's health (0–4)
HISCORE	= 1 if subject scored 75% or better on the quiz
HIAGE	= 1 if subject is 60 years old or older
NOPAIDLEAVE	= 1 if subject is working but does not have paid sick leave
FAILCHECKS	= 1 if subject provided questionable or inconsistent answers in parts of the survey
HIINFORMATION	= 1 if subject acquires health information a few hours per week or more
ASTHMA	= 1 if subject has experienced asthma in past year
BRONCHITIS	= 1 if subject has experienced chronic bronchitis/ emphysema in past year
LUNGINFECTION	= 1 if subject has experienced lung infection (pneumonia, acute bronchitis) in past year
HEARTDISEASE	= 1 if subject has experienced heart disease symptoms in past year
INCOME	Household income before taxes in 1996
PAINRATING	Rating of pain or discomfort on a scale of 0 (no discomfort) to 8 (severe discomfort), summed over all symptoms
FREQILL	Frequency of illness on a scale of 0 (never) to 4 (almost all or all the time), summed over all symptoms
FREQHOME	Frequency of staying home because of illness in past year on a scale of 0 (never) to 4 (almost all or all the time), summed over all symptoms
CH_IC	Alternative specific dummy for the first alternative in each choice set (Initial Condition)
CH_A	Alternative specific dummy for the middle alternative in each choice set (Alternative A)
RHO	Random effects correlation coefficient for subject's eight SP questions

high quiz scores (75% or higher) and no paid sick leave had less noisy responses. The first result is consistent with the intuition that subjects who have digested and understood the health symptom background materials may have better-defined preferences for health–money trade-offs. However, we would have expected subjects with no paid leave to have noisier ratings, everything else equal. The survey told all subjects to assume that their lost ‘sick’ time would be covered by paid

sick leave. For those subjects who do not have paid sick leave in reality, we expected to observe noise in their ratings because the survey assumption conflicts with their real situation. However, because the instructions were different from their actual situation, these subjects may have been more aware of the instructions and thus more consistent in their ratings. Alternatively, NOPAIDLEAVE could be capturing other characteristics that influence noise, such as differences in the

Table 3. Graded-pair ordered-probit panel model

Variable	Coefficient	Normalized	<i>t</i> ratio
Health attributes			
NOSE*LNDAYS	0.2720***	0.6096	4.38
EYE*LNDAYS	0.1610***	0.5432	6.33
FLUTTER*LNDAYS	−0.0862***	0.3953	−3.38
BREATH*LNDAYS	−0.0301	0.4288	−1.34
ACHE*LNDAYS	−0.1426***	0.3615	−5.13
SWELL*LNDAYS	−0.1559***	0.3535	−7.39
PAIN*LNDAYS	−0.0182	0.4360	−0.63
NOLIM	0.9242***	1.0000	6.88
SOMELIM	0.0803***	0.4949	2.52
NOSOC	0.1243**	0.5212	3.75
ATHOME	−0.1026**	0.3854	−2.01
NEEDHELP	−0.2796***	0.2795	−7.22
INHOSP	−0.7466***	0.0000	−6.87
Utility of money function (*SQT COST)			
CONSTANT	−0.021935		−1.22
SCORE	−0.0002413**		−2.18
AGE	0.000392**		2.13
EDUCATION	0.0004650		0.46
SYMPTOMATIC	0.009791**		2.13
VIMPORTANT	−0.003691		−1.61
Scale function			
HISORE	0.1502***		2.69
HIAGE	−0.2579***		−4.09
VIMPORTANT	−0.0003		−0.01
NOPAIDLEAVE	0.1024*		1.70
HIINFORMATION	−0.0423		−0.63
FAILCHECKS	−0.2967***		−5.10
Constants			
ALPHA1	−1.2148		
ALPHA2	−0.1648		
ALPHA3	0.6591		
RHO	0.1260***		5.95
Likelihood ratio Chi-squared		536***	
McFadden <i>R</i> -squared		0.076	
Percent correctly predicted		42.4	
Number of observations		2752	

* Significant at the 10% level or better; ** significant at the 5% level or better; *** significant at the 1% level or better.

type of people who are attracted to jobs that do not offer paid sick leave over jobs that do. Finally, elderly subjects and subjects who failed more consistency checks had noisier responses, as we would expect.

DISCRETE-CHOICE ESTIMATES

As discussed above, because of potential problems and biases associated with using conditional logit choice models to estimate the discrete-choice data, we used RPL in the analysis of the choice data. Table 4 reports the RPL results. Each coefficient is estimated as a distribution defined by two parameters: a mean and a standard deviation. We obtain standard errors for both of these estimates, but report only the significance of each. Therefore, a significant mean parameter indicates that the variable is important in influencing choice, while a significant standard deviation parameter suggests that there is substantial taste heterogeneity across subjects. Although we can construct mean parameters for the omitted categories NOSE and NOLIM, there are no corresponding standard deviation estimates for those variables.

Unlike the graded-pair format, subjects evaluated the choice alternatives relative to their current health, which always was represented by the third alternative across choice sets. The subject's current health is estimated as a function of covariates using the health history data collected in the survey. Thus, in addition to the same health attribute and utility of money variables used in the graded-pair analysis, the discrete-choice RPL estimates include current health covariates and two alternative-specific constants, which indicate the probability of choosing the first or second relative to the third alternative.

We again have effects-coded the health attributes, so the coefficients on the omitted attributes NOSE and NOLIM are reported as the negative sum of the included categories. We interpret statistical significance relative to mean effects for the health attributes. Symptom variables are interacted with duration. Finally, Table 4 also reports both nominal and normalized coefficients.

Symptom was allowed to vary within pairwise comparisons in the graded-pair format but was held constant within choice sets. Because of this difference in the experimental design, we expected to see corresponding differences in estimated symptom utility weights. In particular, we expected to see less

variation among the symptom coefficients compared with the graded pairs. Instead, we observe more variation (the normalized symptom coefficients vary from 0.54 to 0.97) but lower statistical significance relative to the mean effect.ⁱⁱ Moreover, all the normalized choice coefficients are larger than the corresponding pairs coefficients. PAIN is now the worst condition and ACHE now ranked as one of the milder conditions rather than one of the more severe conditions, as indicated in the pairs estimates.

In contrast to the symptom attribute, all activity levels but ATHOME are statistically significant. Furthermore, relative to the normalized endpoints, the choice activity coefficients are larger than the corresponding pairs activity coefficients. In particular, we observe very little difference in the choice NOLIM and SOMELIM activity levels, while there is a large decline in utility between NOLIM and SOMELIM in the pairs estimates. We continue to see NOSOC rated higher than SOMELIM, but the difference again is insignificant.^{jj}

Among the symptom standard deviations, none of the standard deviations for symptoms are significant. Consequently, there is no evidence of preference heterogeneity for the symptom attributes. This result is in striking contrast to the activity standard deviations. All of these are significant and large relative to their corresponding activity categories, indicating considerable variation across observations that are not accounted for elsewhere in the model.

We observe the same pattern of signs in the marginal utility of money function as in the graded-pair model. However, the constant term VIMPOR-TANT and EDUCATION now are significant, while AGE is not. The variable VIMPOR-TANT is intended to capture subjects' attitudes toward health.^{kk} The significant negative coefficient may indicate that subjects who have strong attitudes toward health perceive that they have more personal control over their health, or that there are less expensive alternatives to treatment, leading to lower WTP. Because of differences in scaling, it is not possible to directly compare magnitudes of utility of money parameter differences between the two models.

All four of the respiratory and cardiovascular conditions in the current health function are statistically significant and indicate worse health than overall average health in the sample. ASTHMA and HEARTDISEASE were insignificant when entered independently, but significant

Table 4. Choice random-parameter logit panel model

Variable	Coefficient	Normalized	<i>t</i> ratio
Health attributes			
NOSE*LNDAYS	0.4522**	0.9675	2.12
Estimated SD	—		
EYE*LNDAYS	0.1653**	0.7995	2.07
Estimated SD	0.0001		0.00
FLUTTER*LNDAYS	−0.1851***	0.5944	−2.17
Estimated SD	0.0134		0.02
BREATH*LNDAYS	−0.0786	0.6568	−0.94
Estimated SD	0.0081		0.01
ACHE*LNDAYS	0.1101	0.7672	1.19
Estimated SD	0.0074		0.01
SWELL*LNDAYS	−0.1916*	0.5906	−1.99
Estimated SD	0.1016		0.29
PAIN*LNDAYS	−0.2723***	0.5434	−3.23
Estimated SD	0.0273		0.04
NOLIM	0.5078*	1.0000	1.78
Estimated SD	—		
SOMELIM	0.4864***	0.9875	4.87
Estimated SD	0.3133**		2.10
NOSOC	0.5998***	1.0538	6.06
Estimated SD	0.4752***		3.57
ATHOME	0.0038	0.7050	0.04
Estimated SD	0.5444***		3.82
NEEDHELP	−0.3971***	0.4704	−3.30
Estimated SD	0.5104***		2.59
INHOSP	−1.2007***	0.0000	−6.22
Estimated SD	1.4752***		10.01
Utility of money function (*SQTCOST)			
CONSTANT	−0.1967***		−9.35
SCORE	−0.0000276***		−3.69
AGE	0.000323		1.16
EDUCATION	0.000716***		8.81
SYMPTOMATIC	0.008012*		1.72
VIMPORTANT	−0.011477***		−3.33
Current health function			
ASTHMA*FREQHOME	−0.0953***		−4.39
BRONCHITIS	−0.7115***		−3.21
LUNGINFECTION	−0.2569***		−4.42
HEARTDISEASE*FREQHOME	−0.1335***		−3.98
AGESQUARED	0.0000508***		2.56
INCOME	0.002608***		7.46
PAINRATING	0.003662***		3.72
VIMPORTANT	0.0986***		4.54
FREQILL	−0.1030***		−6.94
FREQHOME	0.2059***		7.14
Alternative-specific constants			
CH_IC	−1.2888***		−4.29
CH_A	0.2423		1.24
Likelihood ratio Chi-squared		557***	
McFadden <i>R</i> -squared		0.093	
Percent correctly predicted		39.3	
Number of observations		2752	

* Significant at the 10% level or better; ** significant at the 5% level or better; *** significant at the 1% level or better.

when interacted with FREQHOME. Thus, ASTHMA and HEARTDISEASE are associated with lower than average health only when they are severe enough to require staying home. Of the four conditions, chronic bronchitis is by far the most serious, with disutility 2.5 times greater than the next most serious condition, while asthma is the least serious.

Age-squared is also positive and highly significant. Generally we expect that older people experience lower levels of health than younger people. It is worth noting, however, that we are modelling subjective health utility, not health *per se*. This result may be interpreted as saying that older people obtain higher levels of utility from the health they have. This is consistent with the positive sign on AGE in the utility of money function, which says that older people value money less than health compared with younger people.

Recall that the experimental design required that cost be ordered from zero to high values among the initial condition IC and alternatives A and B, while activity restrictions are ordered inversely. Thus, some subjects may have tended to choose A simply because it had better health than IC and lower cost than B. CH_A is significant in models where money and current health functions are treated as constants, but insignificant when these functions are included. This result means that including utility of money and current health functions eliminates the middle alternative bias. However, the significant negative coefficient for CH_IC indicates that subjects chose IC relative to A and B less often than the explanatory variables alone would predict.

JOINT GRADED-PAIR, DISCRETE-CHOICE ESTIMATES

Our goal is to obtain estimates suitable for use in a wide range of benefit–cost applications. Recall that the graded-pair format is designed to obtain information on marginal trade-offs among health attributes and cost because there is no unambiguous baseline reference point and no health conditions with zero cost. On the other hand, the discrete-choice format can obtain values for changes in health relative to an identified reference point, but reduces the salience of the symptom attribute because symptom levels do not vary within a given choice set. Consequently, because each SP format has strengths as well as limitations, we pool the data

and estimate a joint model to capture all the available information on subjects' preferences in one set of parameters. Estimating overall preferences in a joint model thus provides a more complete picture of subjects' utility functions for calculating WTP than does either format individually.

Although it would be interesting to estimate a joint panel model, such estimates are beyond the scope of the present study. Instead, we estimate a pooled model using the same specifications as the panel models discussed above, but do not control for intra-subject correlations.¹¹ We then compare the results with the individual panel models.

From Equation (1),

$$U_{st}^i = \lambda_f \cdot V^i(X_{st}, Z^i, P_{st}; \beta, \delta_s^i) + e_{st}^i, \quad (14)$$

where now s indexes an attribute profile in either a graded-pair or choice repetition. We include a choice utility scale function to control for different variances in the two data sets, so $\lambda_f = \lambda_c$ for the choice format and $\lambda_f = 1$ for the graded-pair format.^{mm} The utility parameters β are constrained to be the same for both elicitation formats. However, the utility of money function parameters are estimated separately. Thus,

$$\delta_f^i(Z^i) = \left[\gamma_{f0} + \sum_m (\gamma_{fm} \cdot Z_m^i) \right], \quad (15)$$

where again f indexes the graded-pair and choice formats, respectively. We include the same utility of money variables used in previous models. The joint model also incorporates the current health condition variables used previously, so that WTP values can be estimated relative to different individual current health reference points. Information on current health utility comes only from the choice format. Finally, we combine SOMELIM and NO-SOC into a single category. Previous results indicated that subjects discriminated poorly between these two mild restriction categories.

Figure 3 compares the joint estimates with the discrete-choice and graded-pair panel models. The joint estimates fall between the two panel estimates in nearly every case, although they are clearly not a simple average of the two. The joint estimates track the graded-pair point estimates somewhat more closely than the discrete-choice estimates for the more severe symptoms. Conversely, the joint estimates track the discrete-choice estimates more closely for the activity categories. Thus, the joint estimates have the desired property of giving more weight to the format with the smaller variance for each parameter, i.e. the more precise estimate.

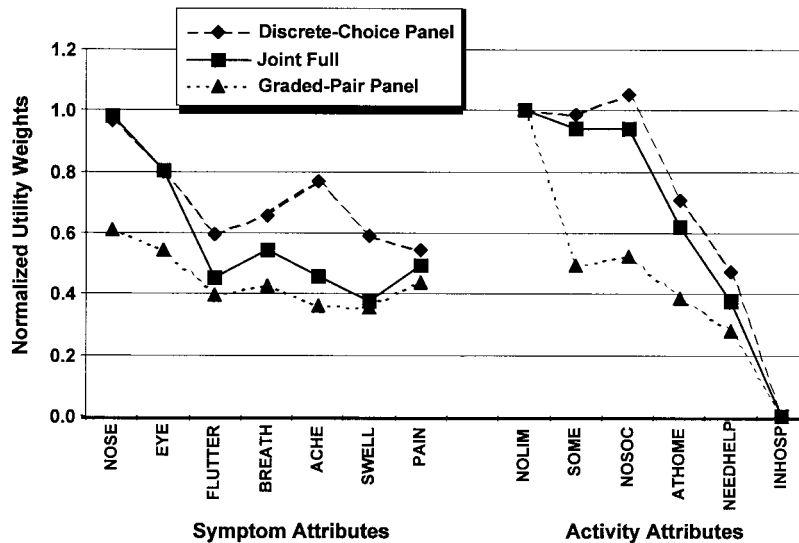


Figure 3. Comparison of joint and individual-panel models

Table 5 reports estimates for the joint model. As in previous models, we present both nominal and normalized coefficients. The pattern of normalized symptom utility weights reflects more variability than either the graded-pair or choice utility weights separately, as also seen in Figure 3. The coefficients for NOSE and EYE are significantly different from the other symptom variables. In addition, SWELL is also significantly different from the other variables. In addition, all variables are significantly different from the mean symptom effect, and only BREATH is significantly different than the mean effect at the 10% level. Thus, the symptom levels in the joint model exhibit substantial variation and more significance than either of the two separate models.

As in the choice model, only ATHOME is statistically insignificant relative to the mean activity effect. However, *t*-tests show that ATHOME is significantly different than the milder category of SOMELIM/NOSOC and activity levels decrease monotonically as daily activity restrictions become more severe. While NOLIM and SOMELIM/NOSOC are significantly different than the mean effect, they are statistically equivalent to each other. Finally, as we observed in Figure 3, the variation in activity levels reflects the choice model more closely than the graded-pair model.

Utility of money functions are estimated separately for the graded-pair and discrete-choice data

to account for systematic differences between the two formats. The implicit marginal utility of money is much smaller in the graded-pair data than the discrete-choice data. There are two possible reasons for the large difference in marginal effects. First, the nature of the discrete-choice task induces subjects to focus on the attribute levels themselves, rather than on differences among the attribute levels. Most people solve the choice valuation problem by eliminating undesirable alternatives based on one or two unattractive attribute levels. Consequently, the task itself simply may cause subjects' to pay more attention to the cost attribute in evaluating alternatives. Second, because symptoms were held constant across two of the three alternatives, subjects did not have to pay as much attention to this attribute as they did in the pairs task. Therefore, they may have had more cognitive energy to focus on the other attributes, including cost. The other utility of money coefficients exhibit similar patterns as in the previous two models.

In the joint model, the relative scale function fixes differences in variance in the two data sets. We use the same scale variables as before, but their interpretation is different in this model. The relative scale function indicates how the variance or noise varies systematically in the choice data set relative to the pairs data set. A positive coefficient indicates a source of lower variance in the choice data. Overall, this function is positive,

Table 5. Joint model

Variable	Coefficient	Normalized	<i>t</i> ratio
Health Attributes			
NOSE*LNDAYS	0.2572***	0.9829	5.52
EYE*LNDAYS	0.1405***	0.8035	7.04
FLUTTER*LNDAYS	−0.0871***	0.4535	−4.59
BREATH*LNDAYS	−0.0294*	0.5422	−1.69
ACHE*LNDAYS	−0.0831***	0.4596	−4.08
SWELL*LNDAYS	−0.1372***	0.3764	−7.91
PAIN*LNDAYS	−0.0609***	0.4938	−3.05
NOLIM	0.2683***	1.0000	4.47
SOMELIM/NOSOC	0.2299***	0.9410	9.39
ATHOME	0.0225	0.6220	0.84
NEEDHELP	−0.1387***	0.3741	−5.36
INHOSP	−0.3820***	0.0000	−9.48
Utility of money function (*SQT COST)			
CH_CONSTANT	−0.1072***		−6.54
CH_SCORE	−0.0001819**		−2.17
CH_AGE	0.000297*		1.65
CH_EDUCATION	0.003968***		5.13
CH_SYMPOMATIC	0.003815		1.03
CH_VIMPORTANT	−0.004019**		−2.09
GP_CONSTANT	−0.007188		−0.40
GP_SCORE	−0.0002305**		−2.22
GP_AGE	0.000432**		2.44
GP_EDUCATION	−0.0005507		−0.53
GP_SYMPOMATIC	0.008877*		1.82
GP_VIMPORTANT	−0.004918**		−2.17
Current health condition			
CH_ASTHMA*FREQHOME	−0.0658***		−2.58
CH_BRONCHITIS	−0.3523**		−2.29
CH_LUNGINFECTION	−0.1679***		−2.53
CH_HEARTDISEASE*FREQHOME	−0.1621***		−2.50
CH_AGESQUARED	0.0000267*		1.62
CH_INCOME	0.001624***		4.39
CH_PAINRATING	0.002615***		2.46
CH_VIMPORTANT	0.0466***		3.27
CH_FREQILL	−0.0496***		−2.85
CH_FREQHOME	0.1124***		3.44
Relative scale function			
CH_HIScore	0.9345***		5.03
CH_HIAGE	−0.8042***		−3.69
CH_VIMPORTANT	0.0718		1.20
CH_NOPAIDLEAVE	−0.1327		−0.77
CH_HIINFORMATION	−0.3505*		−1.80
CH_FAILCHECKS	0.1819		0.89
Alternative-specific constants			
CH_A	−0.5327***		−3.28
CH_B	0.2809**		2.14
GP_ALPHA1	−1.1330		
GP_ALPHA2	−0.1635		
GP_ALPHA3	0.6159		
Likelihood ratio Chi-squared		827***	
McFadden <i>R</i> -squared		0.063	
Graded-pair percent correctly predicted		42.1	
Choice percent correctly predicted		47.5	
Number of observations		5504	

* Significant at 10% level; ** significant at 5% level; *** significant at 1% level.

indicating that the pairs data contain significantly more noise than the discrete-choice data. This result is not surprising because of the more cognitively burdensome nature of the pairs task.

WILLINGNESS-TO-PAY ESTIMATES

Because the joint model captures all the available information on subjects' preferences in one set of parameters, we use this model to estimate WTP. As discussed above, we can convert the parameters of the utility index shown in the joint model into a dollar metric by rescaling the index using the marginal utility of money. Employing SP techniques allows us to recover WTP estimates for all relevant combinations of symptoms and activity levels in the experimental design.ⁿⁿ For example, using Equation (13), the average WTP to avoid 1 day of coughing, wheezing and shortness of breath, which confines subjects to their homes but allows them to care for themselves, is calculated as follows:

$$\begin{aligned} \text{WTP}(\text{BREATH, ATHOME, Days}) &= \frac{\text{Current Utility (Days)} - \text{Utility (BREATH, ATHOME, Days)}}{\text{Marginal utility of money}} \\ &= \frac{\ln(1 + 1) \cdot V_{\text{Current}} - [\ln(1 + 1) \cdot \beta_{\text{BREATH}} + \beta_{\text{ATHOME}}]}{\frac{1}{2} \cdot \left[\frac{-dV}{d\text{Cost}^{\text{CH}}} + \frac{-dV}{d\text{Cost}^{\text{GP}}} \right]} \\ &= \frac{0.6931 \cdot 0.2955 - [0.6931 \cdot (-0.0294) + (0.0225)]}{\frac{1}{2} \cdot \left[\frac{0.01975}{2 \cdot \sqrt{375}} + \frac{0.05859}{2 \cdot \sqrt{202}} \right]} = 158, \end{aligned}$$

where CH and GP refer to choice and graded-pair estimates, respectively, and cost is averaged over all the questions in the each question format where these two symptom and activity levels appear together.^{oo} The bootstrapped 90% confidence interval is CAN\$100–CAN\$225. The confidence interval is obtained by drawing 1000 times from the multivariate normal distribution of coefficients and their variance–covariance matrix. The foregoing calculation is performed for each draw, and the confidence interval lower and upper bounds of the WTP estimates are the 5th and 95th percentile values, respectively.

Table 6 shows WTP estimates and 90% confidence intervals to avoid 1-day, 5-day and 10-day episodes of the relevant symptom and activity level combinations, given subjects' baseline health for the past year. The first column of estimates shows WTP estimates for the MILD activity restriction, which combines SOMELIM and NOSOC. We show zero WTP for 1-day episodes of NOSE, EYE and BREATH. The point estimates for these symptoms are negative, meaning that the estimated average baseline current health in our sample actually was less than the estimated utility for these outcomes. Moving from these health states to current health would reduce welfare, hence the negative point estimate for WTP. This result is not surprising considering that approximately 36% of our 344 subjects have experienced cardiovascular or respiratory conditions within the past year. In addition, non-symptomatic subjects may suffer from a variety of other ailments. Thus, the results show that on average this sample of subjects is not willing to pay to avoid an episode of these mild conditions.

Examining the WTP estimates across symptoms shows that there are essentially two levels of WTP estimates—relatively low numbers associated with the milder symptoms of NOSE and EYE and higher numbers associated with the other five symptoms. These results reflect the variation in the symptom coefficients in the joint estimates. Table 6 also shows that as activity restrictions increase for a given symptom (i.e. as one moves to the right from 'no limitations' to 'in hospital'), WTP also increases. Differences between WTP estimates in adjacent activity levels generally are not statistically significant.

Table 6. WTP estimates for each symptom/activity level combination for 1-day, 5-day and 10-day episodes (1997 CAN\$)

Symptoms	Duration (days)	Daily activity levels			
		MILD ^a	ATHOME	NEEDHELP	INHOSP
NOSE	1	0 ^b (-168/-83)	3 (-50/57)	— ^c	— ^c
	5	0 ^b (-204/4)	29 (-89/148)	— ^c	— ^c
	10	0 ^b (-225/53)	44 (-110/199)	— ^c	— ^c
EYE	1	0 ^b (-119/-32)	54 (-2/108)	— ^c	— ^c
	5	29 (-76/138)	162 (44/282)	— ^c	— ^c
	10	85 (-53/232)	222 (67/379)	— ^c	— ^c
FLUTTER	1	24 (-26/73)	171 (114/236)	288 (232/359)	427 (352/517)
	5	310 (192/438)	468 (341/615)	588 (462/741)	704 (568/870)
	10	468 (309/641)	631 (466/825)	754 (589/955)	857 (684/1065)
BREATH	1	0 ^b (-60/49)	158 (100/225)	286 (224/363)	448 (371/536)
	5	266 (141/405)	435 (299/589)	566 (427/732)	712 (566/872)
	10	415 (249/602)	589 (411/789)	721 (537/937)	857 (668/1058)
ACHE	1	26 (-36/86)	199 (133/275)	335 (267/420)	513 (426/623)
	5	362 (214/523)	544 (382/722)	682 (529/866)	845 (677/1046)
	10	547 (350/762)	734 (524/967)	873 (668/1114)	1027 (813/1278)
SWELL	1	56 (-1/117)	229 (164/306)	365 (295/452)	535 (443/644)
	5	439 (296/598)	621 (469/799)	761 (598/954)	908 (737/1111)
	10	650 (459/869)	837 (634/1073)	979 (767/1228)	1114 (896/1375)
PAIN	1	14 (-48/75)	190 (125/263)	329 (260/411)	510 (426/611)
	5	338 (193/489)	522 (370/693)	663 (508/841)	827 (665/1015)
	10	516 (322/719)	705 (503/934)	848 (645/1084)	1002 (791/1240)

^a MILD combines SOMELIM and NOSOC.^b Negative point estimate shown as zero WTP. See text for interpretation.^c These combinations fall outside the scope of the experimental design.

Overall, WTP estimates show the most dramatic differences across activity levels, and while there are differences among the symptoms, these differences appear to be less salient to subjects for a given activity limitation.

CONCLUSIONS

Using two different SP elicitation formats is a significant improvement over previous health valuation studies. The two formats collect slightly different information and induce subjects to employ different evaluation strategies. Thus, combining both formats allows us to capitalize on the information provided by each format and on the different cognitive processes that subjects use in each situation. The two formats also differ in their formal utility theoretic basis. The graded-pair format elicits values for marginal trade-offs among health states, whereas the discrete-choice format elicits total values to avoid a given condition relative to the subject's current health. Thus, each elicitation format offers certain advantages, and we believe that employing both formats provides more robust and valid estimates.

Comparisons of WTP estimates from the joint model with the few published estimates for similar conditions based on conventional CV methodology indicates our estimates are not systematically larger or smaller [15]. However, unlike other studies in the WTP literature, our SP estimates are derived from complete multi-attribute health state descriptions that are clearly specified and consistent across the same sample of subjects. Thus, we are able to explicitly account for the separate effects of symptoms, duration, activity restrictions, and current health on WTP. The many possible combinations of these attributes correspond to a broad range of cardiovascular and respiratory conditions. This approach allows us to recover consistently derived WTP estimates for these conditions and facilitates matching outcome values more accurately than previously was possible, to evaluate various health care interventions and policies.

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NOTES

- a. Terminology has not been standardized among various disciplines. In this study, we use the term 'stated-preference' to refer to a group of techniques used primarily in market research studies to measure consumer preferences. The term 'conjoint analysis' also has been used to describe some of these techniques. Although CV also could be called an SP technique, CV was developed independently by environmental economists and generally relies on a different set of elicitation formats and analytical approaches.
- b. In the resource economics literature, Gan and Luzar [23] use SP to value hunting trips in Louisiana. Mackenzie [18] values hunting trips in Delaware using SP analysis. Opaluch *et al.* [24] also use SP to describe public preferences for siting a noxious facility. Adamowicz *et al.* [25,26] use SP to explain recreational site choice selection. Johnson *et al.* [27] use SP to estimate electric customers' WTP for environmental and other attributes of electricity generation. Roe *et al.* [28] use SP to value the effects on sport fishing of implementing alternative management plans to restore runs of Atlantic salmon in Maine.
- c. Preference elicitation for assessing welfare changes requires that subject responses conform to basic postulates of utility theory. Among other things, this rules out lexicographic preferences, where subjects refuse attribute trade-offs. SP data often facilitate tests of conformity with utility theoretic requirements [29]. Failure to accept trade-offs often arises because the range of attribute levels offered is not salient. Ensuring appropriate attribute ranges is a basic task of survey development and testing.
- d. Defining the properties of such preferences has been explored by multi-attribute utility theory [30]. For an application to health state utility, see Torrance *et al.* [31].
- e. A third format, *ranking*, has also been used in market research. For a more extensive review of experimental design possibilities, see Green *et al.* [32].

- f. For a review of choice experiments in marketing applications, see Carson *et al.* [33].
- g. This approach has been used in modelling travel behaviour, evaluating new products and estimating recreation demands. See, for example, Louviere and Hensher [34]. Health studies that use this approach include Bryan *et al.* [9], Propper [11] and Ryan and Hughes [8].
- h. See also Huber *et al.* [13], Huber [35], Tversky *et al.* [36], Payne [37,38], Huber and Klein [39], Elrod *et al.* [40], Akaah [41] and Huber and Holbrook [42] for discussions of the different cognitive processes involved in graded-pair versus choice tasks.
- i. We initially treated the QWB function states as separate attributes, but we found in pre-testing that subjects tended to view these function states as interdependent, which complicates the development of an orthogonal experimental design.
- j. The range of cost values offered do not determine higher or lower WTP values. Rather, variations in cost induce trade-offs among attributes. These trade-offs reveal information about marginal utilities, including the marginal utility of money. Calculating WTP using estimated parameters is discussed below.
- k. To avoid introducing correlation into the design by simply deleting combinations that violated constraints, we constructed a series of orthogonal designs for groups of constrained attribute levels, then concatenated the sub-designs into a candidate design from which a design of the desired size and efficiency was drawn.
- l. Thus, NOSE and EYE are never seen with ATH-OME, NEEDHELP and INHOSP, as well as values of \$200 and \$500.
- m. The payment vehicle for both formats is described as illness-related costs that are not covered by the government health system or a company insurance plan. These costs are associated with items that reduce discomfort or the length of illness (such as vitamins, medicines, air filters or humidifiers, special foods or liquids, or other optional treatments). These costs may also include such costs as child care, while sick, or transportation to the doctor. Subjects were instructed to assume that any missed time from work would be covered by paid sick leave.
- n. The choice format was added to the study after a large-scale pre-test of the graded-pairs format. Peer reviewers suggested that adding a graded-pairs component to the study would provide better ability to control for current health status, as well as offer the benefits of multiple-format elicitations discussed above. For details on the evolution of the experimental design, see Johnson *et al.* [15,44] and Desvousges *et al.* [43].
- o. These illnesses included minor lung illnesses, lung infections, chronic lung conditions, coronary artery disease, heart failure and heart rhythm problems.
- p. The computer format was an advantage in this part of the survey because if subjects answered the quiz question incorrectly, the computer provided them with the correct response.
- q. Although we did not actively recruit subjects who had ever experienced the heart and lung illnesses of interest, 43.6% of the sample used in the analysis have been diagnosed with one of these conditions at some point in their lives.
- r. The first social science application of this approach was by McKelvey and Zavoina [45] (see Elrod *et al.* [40], Akaah [41] and Huber and Holbrook [42]). Roe *et al.* [28] and Johnson and Desvousges [46] discuss parameter estimation and WTP calculations for graded-pairs data.
- s. See Keeney and Raiffa [30] for an analysis of the properties of such utility functions.
- t. The square root form was chosen over linear or log by estimating a Box-Cox model.
- u. This procedure assumes that subjects have no systematic preference for screen location.
- v. Because the original response scale indicates which profile is preferred and how much it is preferred, this rearrangement maps response 3 into 5, 2 into 6 and 1 into 7. Response 7 indicates maximum utility difference and 4 indicates indifference; so C'_i equals the recoded response minus 4. We also have estimated binary models (not reported here), which suppress the intensity of preference information. Such models fit the data poorly relative to the ordered probit specification. Also, it is not clear how to classify indifference in a binary model.
- w. The maximum likelihood procedure used to estimate the model parameters normalizes the α_0 threshold at $-\infty$ and α_5 at $+\infty$, and does not include an intercept term.
- x. The way to generalize μ in a probit-type model is to retain the normalization for the base category that $\mu_0^i = 1$. For all other categories, $\mu_k^i = \mu^i$ is modelled as a function of individual characteristics. This guarantees that the estimated value of μ will be positive for every observation and that α_k adjusts to account for the normalization of the base category. Since ρ is the normalized intra-subject correlation, we have controlled for both intra-subject and inter-subject heteroskedasticity, which removes bias in estimated standard errors from both sources. Our ability to separately identify these two sources of heteroskedasticity requires that we assume independence in of the permanent and transitory components of the error term. If these errors are correlated, then the intra-subject and inter-subject variances are confounded to an unknown extent.
- y. This specification is a generalization of Butler and Moffitt's [47] derivation for binary probit.
- z. The basic exposition of the properties of this model can be found in McFadden [48].

- aa. Preliminary estimates were obtained from multinomial probit, which yielded very similar results. We report conditional logit estimates to facilitate subsequent comparisons with random parameter logit estimates. It is currently impractical to estimate random parameter multinomial probit models.
- bb. Because utility is a non-linear function of price, WTP is a function of some assumed price level. The WTP calculations reported below incorporate average price combinations that occur with different symptom/activity levels.
- cc. Effects codes are an alternative to dummy variables for describing and modelling qualitative variables. For example, a two-level attribute is coded $(-1, 1)$, a three-level attribute is coded $(-1, -1, 01, 10)$, etc. Thus, the coefficient for the omitted category is the negative sum of the coefficients for the included categories. Similarly, the standard error for the omitted category coefficient is derived from the variance-covariance matrix of the included categories. *t*-tests of significance indicate whether a coefficient is different from the variable's mean effect, which is equal to zero by definition. For dummy variables, significance tests are relative to all omitted categories, which are coded as zero. Effects coding avoids the problem of evaluating all coefficients relative to a combination of omitted categories (see Louviere [49]).
- dd. The transformation $\ln(\text{duration} + 1)$ allows for diminishing marginal utility of duration.
- ee. All analysis for this study was performed using Aptech System's GAUSS software.
- ff. See Johnson *et al.* [15] for comparisons with non-panel ordered-probit estimates.
- gg. Symptom was held constant within choice sets in the choice questions, so it is necessary to interact duration with symptom in the choice models. Although a main-effects experimental design generally does not allow modelling interaction effects, the maximum correlation with other explanatory variables is only 0.5. Much of this correlation was induced by eliminating implausible combinations, rather than by the design itself. This level of correlation generally is acceptable in empirical work and was not serious enough to cause insignificant coefficients (see Judge *et al.* [50], p. 868).
- hh. All tests of significance are performed with *t*-tests.
 - ii. In addition, NOSE and EYE are significantly different than FLUTTER, PAIN, BREATH and SWELL, but there are no other significant differences among the symptoms.
 - jj. NOSOC is also insignificantly different from NOLIM.
- kk. Subjects were asked a series of four questions at the beginning of the survey on their opinions of how various factors (i.e. heredity, smoking, drinking and exercise) influenced health. VIMPOR-TANT is an index of the number of factors subjects rated as 'very important' in influencing health.
- ll. Note that the data from the two formats are not literally stacked. Rather, the joint likelihood function is the sum of the likelihood functions for the separate models, with some parameters constrained to be the same in the two functions. See Adamowicz *et al.* [25,26] and Hensher *et al.* [51] for examples of this type of joint estimation.
- mm. The joint model omits the within-data set scale parameters, which had negligible effects on other parameters and do not enter into WTP calculations.
- nn. Some combinations of attribute levels, such as having a stuffy nose and sore throat and being in hospital, are not plausible and thus WTP estimates for these combinations are not presented because they fall outside the range of the experimental design.
- oo. The estimate of V_{Current} is obtained by evaluating the current health function in the joint model at the sample means of the covariates and summing across these covariates. Likewise, the utility of money functions for the graded-pair and choice formats are evaluated at the covariate sample means and summed.

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